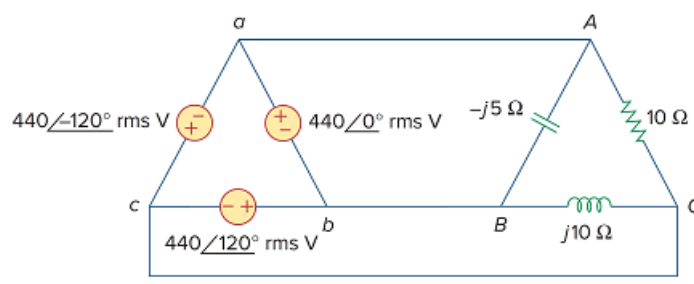


Problem

Find the line currents in the unbalanced three-phase circuit of Fig. 12.26 and the real power absorbed by the load.

Figure 12.26



Step-by-step solution

Step 1 of 7

Refer to Figure 12.26 in the text book.

Notice that each phase impedances is connected in parallel to their respective line voltage sources.

Calculate the value of the current in the branch A-B.

$$\begin{aligned} I_{AB} &= \frac{440\angle 0^\circ}{-j5} \\ &= \frac{440\angle 0^\circ}{5\angle -90^\circ} \\ &= 88\angle 90^\circ \text{ A rms} \end{aligned}$$

Therefore, the value of the current I_{AB} is $88\angle 90^\circ$ A rms.

Step 2 of 7

Calculate the value of the current in the branch B-C.

$$\begin{aligned} I_{BC} &= \frac{440\angle 120^\circ}{j10} \\ &= \frac{440\angle 120^\circ}{10\angle 90^\circ} \\ &= 44\angle 30^\circ \text{ A rms} \end{aligned}$$

Therefore, the value of the current I_{BC} is $44\angle 30^\circ$ A rms.

Step 3 of 7

Calculate the value of the current in the branch C-A.

$$\begin{aligned} I_{CA} &= \frac{440\angle -120^\circ}{10} \\ &= \frac{440\angle -120^\circ}{10\angle 0^\circ} \\ &= 44\angle -120^\circ \text{ A rms} \end{aligned}$$

Therefore, the value of the current I_{CA} is $44\angle -120^\circ \text{ A rms}$.

Step 4 of 7

Calculate the value of the line current I_{aA} .

$$I_{aA} = I_{AB} - I_{CA}$$

Substitute $88\angle 90^\circ \text{ A rms}$ for I_{AB} and $44\angle -120^\circ \text{ A rms}$ for I_{CA} .

$$\begin{aligned} I_{aA} &= 88\angle 90^\circ - 44\angle -120^\circ \\ &= 128.01\angle 80.1^\circ \text{ A rms} \end{aligned}$$

Therefore, the value of the line current I_{aA} is $128.01\angle 80.1^\circ \text{ A rms}$.

Step 5 of 7

Calculate the value of the line current I_{bB} .

$$I_{bB} = I_{BC} - I_{AB}$$

Substitute $44\angle 30^\circ \text{ A rms}$ for I_{BC} and $88\angle 90^\circ \text{ A rms}$ for I_{AB} .

$$\begin{aligned} I_{bB} &= 44\angle 30^\circ - 88\angle 90^\circ \\ &= 76.21\angle -60^\circ \text{ A rms} \end{aligned}$$

Therefore, the value of the line current I_{bB} is $76.21\angle -60^\circ \text{ A rms}$.

Step 6 of 7

Calculate the value of the line current I_{cC} .

$$I_{cC} = I_{CA} - I_{BC}$$

Substitute $44\angle 30^\circ \text{ A rms}$ for I_{BC} and $44\angle -120^\circ \text{ A rms}$ for I_{CA} .

$$\begin{aligned} I_{cC} &= 44\angle -120^\circ - 44\angle 30^\circ \\ &= 85\angle -135^\circ \text{ A rms} \end{aligned}$$

Therefore, the value of the line current I_{cC} is $85\angle -135^\circ \text{ A rms}$.

Step 7 of 7

The real power absorbed by the load is due to the resistance present in the load circuit.

Calculate the value of the real power absorbed by the load.

$$P = |I_{Cd}|^2 (10)$$

Substitute 44 A rms for $|I_{Cd}|$.

$$\begin{aligned} P &= (44)^2 (10) \\ &= (1936)(10) \\ &= 19.36 \text{ kW} \end{aligned}$$

Therefore, the value of the real power P absorbed by the load is 19.36 kW.